



# SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



|                                                                                                                   |                           |
|-------------------------------------------------------------------------------------------------------------------|---------------------------|
| LOCATION: Train 5                                                                                                 | AREA: E                   |
| PERMIT TO WORK No.: -                                                                                             | TYPE OF SCAFFOLD: Hanging |
| SCAFFOLD DRAWING No: PMS-WA-ARCO-042                                                                              |                           |
| RISK CLASSIFICATION: <input checked="" type="radio"/> High <input type="radio"/> Medium <input type="radio"/> Low |                           |

## SCAFFOLD TO BE INSPECTED BY PIC BEFORE TAGGING

### BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of a Hanging scaffold required for Leak Detection PoC camera installation. under work order 4071323-514UZ-776 at Train 5. The scaffold envelope is 2.4m x 2.1m x 3m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, and support/tie arrangements as indicated on the approved drawing. The structure is 10m above ground level.

**SAFE WORKING LOAD (SWL): 1.5 kN/m<sup>2</sup>**

**Total Horizontal Load in the X Direction ( $F_X$ ): 0.335 kN**

**Total Horizontal Load in the Z Direction ( $F_Z$ ): 6.59 kN**

**Total Vertical Load in the Y Direction ( $F_Y$ ): 16.864 kN**

| Role        | ID          | Name               | Signature | Date |
|-------------|-------------|--------------------|-----------|------|
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|                              |             |                        |                       |
|------------------------------|-------------|------------------------|-----------------------|
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|------------------------------|-------------|------------------------|-----------------------|

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|                                                                                   |                  |                        |                               |                                                                                     |
|-----------------------------------------------------------------------------------|------------------|------------------------|-------------------------------|-------------------------------------------------------------------------------------|
|  | Project Name:    | SCAFFOLD DESIGN REPORT |                               |  |
|                                                                                   | Scaffold Type:   | Hanging                | Calculation Sheet             |                                                                                     |
|                                                                                   | Document Number: | PMS-WA-ARCO-042        | Prepared By: Nnamani, Ifeanyi |                                                                                     |

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ENGINEERING DRAWINGS

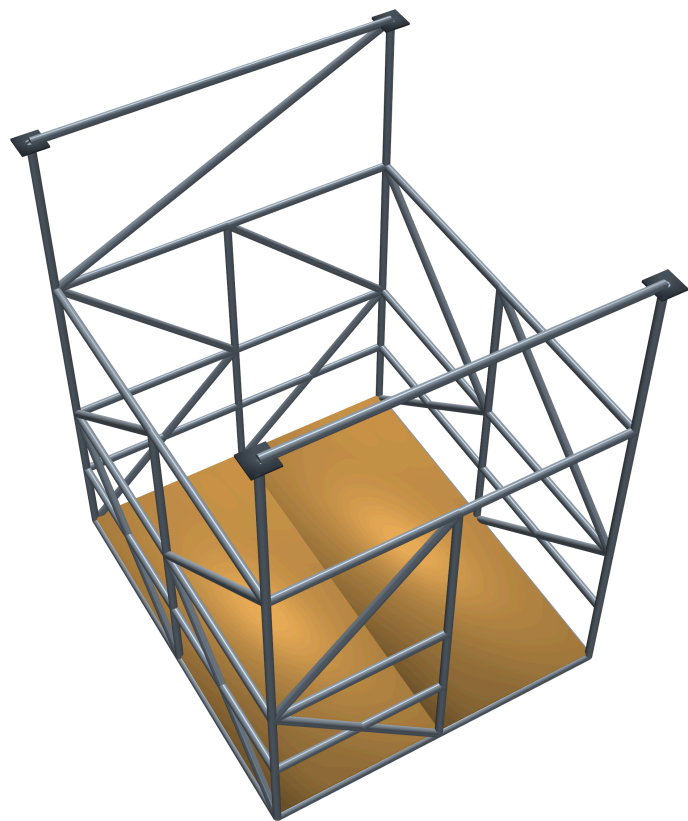
Engineering Drawing - Front View

Engineering Drawing - Side View

Engineering Drawing - Plan View

Engineering Drawing - 3D View

3D ISOMETRIC VIEW



|                                                                                   |                  |                               |                                      |                                                                                     |
|-----------------------------------------------------------------------------------|------------------|-------------------------------|--------------------------------------|-------------------------------------------------------------------------------------|
|  | Project Name:    | <b>SCAFFOLD DESIGN REPORT</b> |                                      |  |
|                                                                                   | Scaffold Type:   | Hanging                       | Calculation Sheet                    |                                                                                     |
|                                                                                   | Document Number: | PMS-WA-ARCO-042               | Prepared By: <b>Nnamani, Ifeanyi</b> |                                                                                     |

## GENERAL NOTE

**BS EN 12811-1, 4.2.1.2 / 10.3.2.2**

Scaffold tube material has a minimum nominal yield strength of 235 N/mm<sup>2</sup> (BS EN 12811-1, 4.2.1.2). A partial safety factor  $\gamma_M = 1.1$  (BS EN 12811-1, 10.3.2.2) has been applied, giving a design resistance of 213.6 N/mm<sup>2</sup> used in member capacity checks. This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

## Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

## Modelling Philosophy

**BS EN 12811-1, 10.2.3.2 / 10.2.3.3**

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ( $K_{FX}/K_{FZ} = 5.596$  kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot.

## LOAD CASES

### Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m<sup>3</sup>**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

### Live Load on Platforms

**BS EN 12811-1**

Platform live loading is applied to the scaffold working platforms as member UDLs in the STAAD model. The load intensity is multiplied by the tributary width of each supporting member to obtain the applied member line load. (LC 2).

| Member                             | Tributary Width (m)     | Load Intensity (kN/m <sup>2</sup> ) | Member Load (kN/m) |
|------------------------------------|-------------------------|-------------------------------------|--------------------|
| <b>Members 2, 4, 11, 12 (Edge)</b> | 1.200 / 2 = 0.600       | 1.500                               | <b>0.900</b>       |
| <b>Members 17, 18 (Interior)</b>   | (0.600 + 0.600) = 1.200 | 1.500                               | <b>1.800</b>       |

*Load Intensity = 1.50 kN/m<sup>2</sup> — BS EN 12811-1 Class 2*

Member Load = Load Intensity × Tributary Width. Edge members carry half the adjacent bay width; interior members carry half the bay from each side.

Governing load intensity (SWL) = **1.5 kN/m<sup>2</sup>** — BS EN 12811-1 Class 2.

Total applied platform live load = **7.56 kN**.

### Handrail Load

Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **0.95 kN** (member UDL 0.15 kN/m)

Handrail Load (HLZ) = **1.44 kN** (member UDL 0.15 kN/m)

## Wind Load

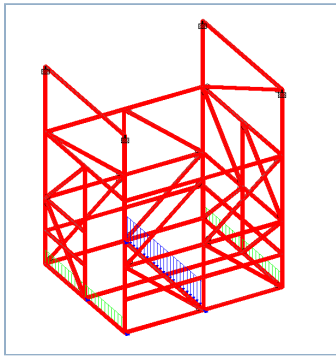
EN 1991-1-4:2005

Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.071567 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

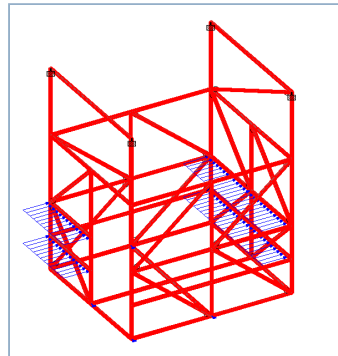
Total Wind Load - X Axis (WLX) = **2.91 kN**

Total Wind Load - Z Axis (WLZ) = **2.95 kN**

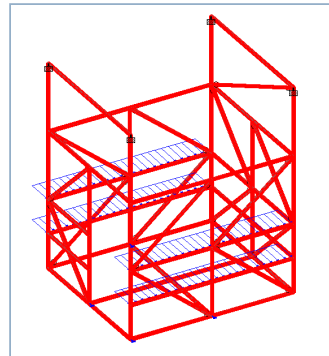
## Load Diagrams



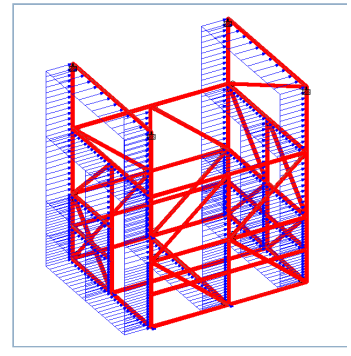
Live Load (LL)



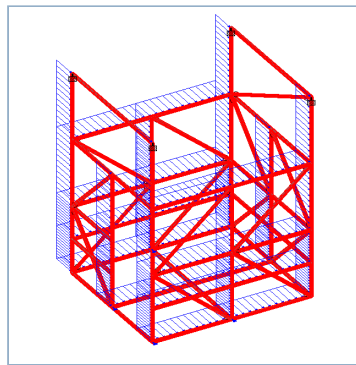
Handrail Load X (HLX)



Handrail Load Z (HLZ)



Wind Load X (WLX)



Wind Load Z (WLZ)

## WIND LOAD CALCULATION - EN 1991-1-4:2005

| Reference              | Calculation                                                                             | Value        | Unit |
|------------------------|-----------------------------------------------------------------------------------------|--------------|------|
|                        | Basic wind speed (3-second gust)                                                        |              |      |
|                        | $V_3 =$                                                                                 | <b>36.0</b>  | m/s  |
|                        | <i>Eurocode <math>V_{b,0}</math> is the characteristic 10-minute mean wind velocity</i> |              |      |
| Durst, C.S. (1960)     | Convert 3-second gust to 10-minute mean wind velocity                                   |              |      |
|                        | $V_3 / V_{3600} =$                                                                      | 1.52         |      |
|                        | $V_{600} / V_{3600} =$                                                                  | 1.06         |      |
|                        | $V_{600} = (V_{600}/V_{3600}) \times (V_{3600}/V_3) \times V_3 =$                       | <b>25.11</b> | m/s  |
|                        | Basic wind speed $V_b = V_{600} =$                                                      | <b>25.11</b> | m/s  |
| EN 1991-1-4:2005       | Roughness Length ( $z_0$ ) - Terrain Category 0                                         | 0.003        | m    |
|                        | Terrain category II ( $z_{0,ii}$ )                                                      | 0.05         |      |
| EN 1991-1-4, Table 4.1 | $z_{min}$                                                                               | 1.0          | m    |
|                        | <b>z - height of structure under consideration</b>                                      | <b>13.0</b>  | m    |
| EN 1991-1-4, Sec 4.3   | $C_{r0}(z)$ - Roughness factor                                                          | 1.0          |      |

| Reference                                                 | Calculation                                                             | Value            | Unit              |
|-----------------------------------------------------------|-------------------------------------------------------------------------|------------------|-------------------|
| EN 1991-1-4, Sec 4.3                                      | $C_o(z)$ - Orography factor                                             | 1.0              |                   |
| EN 1991-1-4, Sec 4.3                                      | Terrain factor $K_r = 0.19 \times [z_0/z_{0,ii}]^{0.07}$                | 0.156            |                   |
| EN 1991-1-4, Table 4.1                                    | $z_{min} < z \rightarrow$ Case 1 applies                                |                  |                   |
| <b>Wind Turbulence</b>                                    |                                                                         |                  |                   |
|                                                           | $C_r(z) = K_r \times \ln(z/z_0)$                                        | 1.3067           |                   |
|                                                           | Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$                | 32.804           | m/s               |
| EN 1991-1-4, Sec 4.4                                      | K1                                                                      | 1.0              |                   |
|                                                           | Turbulence intensity $I_v(z) = k1 / [C_o(z) \times \ln(z/z_0)]$         | 0.1194           |                   |
| <b>Peak Velocity Pressure</b>                             |                                                                         |                  |                   |
|                                                           | Density of air rho                                                      | 1.25             | kg/m <sup>3</sup> |
| EN 1991-1-4, Sec 4.5                                      | $q_p(z) = [1 + 7 \times I_v(z)] \times 0.5 \times \rho \times V_m(z)^2$ | <b>1234.7675</b> | N/m <sup>2</sup>  |
| <b>Wind Load = <math>C_f \times q_p \times A_f</math></b> |                                                                         |                  |                   |
| $C_f =$                                                   |                                                                         | 1.2              |                   |
| $q_p =$                                                   |                                                                         | 1.234767         | kN/m <sup>2</sup> |
| $A_f =$                                                   | Tube outer diameter (48.3 mm)                                           | 0.0483           | m                 |
| <b>Wind Load UDL =</b>                                    |                                                                         | <b>0.071567</b>  | <b>kN/m</b>       |

## LOAD COMBINATIONS

### BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**;  
SLS (characteristic) combinations use gamma = **1.0**.

**DL** = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

| LC No. | Combination                         | Component Loadings & Factors                  |
|--------|-------------------------------------|-----------------------------------------------|
| 7      | SLS DL + LL                         | LC1 x 1.0 + LC2 x 1.0                         |
| 8      | SLS DL + LL + HLX                   | LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0             |
| 9      | SLS DL + LL + HLZ                   | LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0             |
| 10     | SLS DL + WLX                        | LC1 x 1.0 + LC5 x 1.0                         |
| 11     | SLS DL + WLZ                        | LC1 x 1.0 + LC6 x 1.0                         |
| 12     | SLS DL + LL + WLX                   | LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0             |
| 13     | SLS DL + LL + WLZ                   | LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0             |
| 14     | SLS DL + LL + WLZ + HLX             | LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC3 x 1.0 |
| 15     | SLS DL + LL + WLZ + HLZ             | LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC4 x 1.0 |
| 16     | ULS 1.5DL + 1.5LL                   | LC1 x 1.5 + LC2 x 1.5                         |
| 17     | ULS 1.5DL + 1.5LL + 1.5HLX          | LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5             |
| 18     | ULS 1.5DL + 1.5LL + 1.5HLZ          | LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5             |
| 19     | ULS 1.5DL + 1.5LL + 1.5WLX          | LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5             |
| 20     | ULS 1.5DL + 1.5LL + 1.5WLZ          | LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5             |
| 21     | ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLX | LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC3 x 1.5 |
| 22     | ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLZ | LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC4 x 1.5 |
| 23     | ULS 1.5DL + 1.5WLX                  | LC1 x 1.5 + LC5 x 1.5                         |
| 24     | ULS 1.5DL + 1.5WLZ                  | LC1 x 1.5 + LC6 x 1.5                         |
| 25     | ULS DL + 1.5WLX                     | LC1 x 1.0 + LC5 x 1.5                         |
| 26     | ULS DL + 1.5WLZ                     | LC1 x 1.0 + LC6 x 1.5                         |

## STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

**Maximum Utilization Ratio = 0.453** at Member No. **18**, Load Combination **22**, Clause **EC-6.2.9.1**. **PASS** - All members adequate; UC ratio below unity.

| Member | Section  | Status | Clause       | UC Ratio | L/C | F <sub>x</sub> (kN) | Type |
|--------|----------|--------|--------------|----------|-----|---------------------|------|
| 18     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.453    | 22  | 0.07                | C    |
| 25     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.388    | 22  | 1.91                | T    |
| 17     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.355    | 19  | 0.00                | C    |
| 14     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.309    | 17  | 0.72                | T    |
| 49     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.307    | 22  | 4.73                | T    |
| 13     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.293    | 16  | 0.50                | T    |
| 26     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.262    | 19  | 1.29                | T    |
| 52     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.236    | 22  | 3.33                | T    |
| 9      | PIP483.6 | PASS   | EC-6.2.9.1   | 0.231    | 22  | 1.17                | T    |
| 24     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.221    | 17  | 1.91                | T    |
| 4      | PIP483.6 | PASS   | EC-6.2.5     | 0.199    | 22  | 0.00                | C    |
| 23     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.193    | 22  | 1.61                | T    |
| 60     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.159    | 22  | 0.92                | T    |
| 37     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.144    | 21  | 2.63                | T    |
| 64     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.138    | 22  | 0.45                | C    |
| 34     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.122    | 22  | 3.31                | T    |
| 48     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.122    | 22  | 0.13                | T    |
| 2      | PIP483.6 | PASS   | EC-6.2.9.1   | 0.114    | 22  | 0.97                | T    |
| 43     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.111    | 22  | 0.70                | C    |
| 12     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.109    | 19  | 0.38                | T    |
| 33     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.107    | 22  | 0.29                | C    |
| 28     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.104    | 22  | 0.69                | C    |
| 8      | PIP483.6 | PASS   | EC-6.2.9.1   | 0.103    | 22  | 1.47                | T    |
| 69     | PIP483.6 | PASS   | EC-6.3.3-661 | 0.100    | 22  | 2.78                | C    |
| 42     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.097    | 22  | 0.12                | C    |
| 5      | PIP483.6 | PASS   | EC-6.2.9.1   | 0.094    | 19  | 1.27                | T    |
| 27     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.094    | 22  | 0.27                | C    |
| 40     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.093    | 17  | 1.99                | T    |
| 6      | PIP483.6 | PASS   | EC-6.2.9.1   | 0.092    | 19  | 1.71                | T    |
| 11     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.089    | 19  | 0.28                | T    |
| 59     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.088    | 22  | 0.94                | C    |
| 30     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.087    | 22  | 0.71                | C    |
| 41     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.085    | 18  | 1.91                | T    |
| 63     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.081    | 22  | 0.09                | C    |
| 45     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.079    | 21  | 0.31                | T    |
| 22     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.078    | 17  | 0.87                | T    |
| 29     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.066    | 18  | 0.18                | C    |
| 38     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.066    | 17  | 1.46                | T    |
| 58     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.065    | 22  | 0.26                | T    |
| 57     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.064    | 22  | 1.03                | C    |
| 31     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.063    | 16  | 0.58                | C    |
| 61     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.061    | 22  | 0.30                | T    |
| 62     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.057    | 19  | 0.31                | C    |

| Member                                                                       | Section  | Status | Clause       | UC Ratio | L/C | F <sub>x</sub> (kN) | Type |
|------------------------------------------------------------------------------|----------|--------|--------------|----------|-----|---------------------|------|
| 32                                                                           | PIP483.6 | PASS   | EC-6.3.3-662 | 0.055    | 18  | 0.15                | C    |
| 55                                                                           | PIP483.6 | PASS   | EC-6.2.9.1   | 0.054    | 22  | 0.60                | T    |
| 44                                                                           | PIP483.6 | PASS   | EC-6.3.3-662 | 0.051    | 16  | 0.11                | C    |
| 39                                                                           | PIP483.6 | PASS   | EC-6.2.9.1   | 0.050    | 21  | 2.20                | T    |
| 47                                                                           | PIP483.6 | PASS   | EC-6.2.9.1   | 0.048    | 17  | 0.27                | T    |
| 21                                                                           | PIP483.6 | PASS   | EC-6.2.9.1   | 0.046    | 22  | 1.60                | T    |
| 56                                                                           | PIP483.6 | PASS   | EC-6.2.9.1   | 0.046    | 22  | 0.60                | T    |
| Showing highest 50 utilization ratios; 21 lower-utilization members omitted. |          |        |              |          |     |                     |      |

## SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **2.780 kN** (Member 69, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class A** (capacity = 10.0 kN).

Check: 2.780 kN ≤ 10.0 kN → **PASS**

Class A right-angle couplers are adequate.

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

| Coupler Type                 | Resistance                                      | Characteristic Value |         |
|------------------------------|-------------------------------------------------|----------------------|---------|
|                              |                                                 | Class A              | Class B |
| Right-angle Coupler (RA)     | Slipping force F <sub>s,k</sub> (kN)            | 10.0                 | 15.0    |
|                              | Cruciform bending moment M <sub>B,k</sub> (kNm) | -                    | 0.8     |
|                              | Pull-apart force F <sub>p,k</sub> (kN)          | 20.0                 | 30.0    |
|                              | Rotational moment M <sub>T,k</sub> (kNm)        | -                    | 0.13    |
| Swivel Coupler (SW)          | Slipping force F <sub>s,k</sub> (kN)            | 6.0                  | 9.0     |
|                              | Cruciform bending moment M <sub>B,k</sub> (kNm) | -                    | 0.8     |
|                              | Pull-apart force F <sub>p,k</sub> (kN)          | 15.0                 | 22.5    |
|                              | Rotational moment M <sub>T,k</sub> (kNm)        | -                    | 0.08    |
| Putlog / Single Coupler (PL) | Slipping force F <sub>s,k</sub> (kN)            | -                    | 3.0     |
|                              | Cruciform bending moment M <sub>B,k</sub> (kNm) | -                    | 0.6     |

Member Axial Force Summary (Top 30 Horizontal Members — ULS Code Check)

| Member | Type | Status | Clause       | UC Ratio | L/C | Axial Force (kN) |
|--------|------|--------|--------------|----------|-----|------------------|
| 69     | C    | PASS   | EC-6.3.3-661 | 0.100    | 22  | 2.780            |
| 70     | C    | PASS   | EC-6.3.3-662 | 0.034    | 18  | 1.950            |
| 74     | C    | PASS   | EC-6.3.3-662 | 0.033    | 19  | 1.570            |
| 79     | C    | PASS   | EC-6.3.3-662 | 0.033    | 19  | 1.570            |
| 67     | C    | PASS   | EC-6.3.3-662 | 0.025    | 19  | 1.270            |
| 76     | C    | PASS   | EC-6.3.3-662 | 0.028    | 18  | 1.230            |
| 57     | C    | PASS   | EC-6.3.3-662 | 0.064    | 22  | 1.030            |
| 73     | C    | PASS   | EC-6.3.3-662 | 0.025    | 22  | 1.020            |
| 2      | T    | PASS   | EC-6.2.9.1   | 0.114    | 22  | 0.970            |
| 3      | T    | PASS   | EC-6.2.9.1   | 0.025    | 22  | 0.960            |
| 59     | C    | PASS   | EC-6.3.3-662 | 0.088    | 22  | 0.940            |
| 66     | C    | PASS   | EC-6.3.3-662 | 0.020    | 20  | 0.930            |

| Member | Type | Status | Clause       | UC Ratio | L/C | Axial Force (kN) |
|--------|------|--------|--------------|----------|-----|------------------|
| 60     | T    | PASS   | EC-6.2.9.1   | 0.159    | 22  | 0.920            |
| 14     | T    | PASS   | EC-6.2.9.1   | 0.309    | 17  | 0.720            |
| 30     | C    | PASS   | EC-6.3.3-662 | 0.087    | 22  | 0.710            |
| 43     | C    | PASS   | EC-6.3.3-662 | 0.111    | 22  | 0.700            |
| 28     | C    | PASS   | EC-6.3.3-662 | 0.104    | 22  | 0.690            |
| 31     | C    | PASS   | EC-6.3.3-662 | 0.063    | 16  | 0.580            |
| 77     | C    | PASS   | EC-6.3.3-662 | 0.018    | 23  | 0.540            |
| 13     | T    | PASS   | EC-6.2.9.1   | 0.293    | 16  | 0.500            |
| 64     | C    | PASS   | EC-6.3.3-662 | 0.138    | 22  | 0.450            |
| 1      | C    | PASS   | EC-6.3.3-662 | 0.016    | 23  | 0.430            |
| 12     | T    | PASS   | EC-6.2.9.1   | 0.109    | 19  | 0.380            |
| 71     | C    | PASS   | EC-6.3.3-662 | 0.013    | 24  | 0.380            |
| 45     | T    | PASS   | EC-6.2.9.1   | 0.079    | 21  | 0.310            |
| 62     | C    | PASS   | EC-6.3.3-662 | 0.057    | 19  | 0.310            |
| 61     | T    | PASS   | EC-6.2.9.1   | 0.061    | 22  | 0.300            |
| 33     | C    | PASS   | EC-6.3.3-662 | 0.107    | 22  | 0.290            |
| 11     | T    | PASS   | EC-6.2.9.1   | 0.089    | 19  | 0.280            |
| 27     | C    | PASS   | EC-6.3.3-662 | 0.094    | 22  | 0.270            |

## DEFLECTION CHECK RESULTS

**BS EN 12811-1, Cl 6.3** Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results. Displacement tables are shown below.

### Vertical Deflection Check

| Parameter                                                                    | Value   | Allowable                            | Check |
|------------------------------------------------------------------------------|---------|--------------------------------------|-------|
| Max vertical displacement - LC 15<br>Member 17   Member length L = 1050.0 mm | 0.96 mm | $L/100 = 1050/100 = 10.5 \text{ mm}$ | PASS  |

### Vertical Displacement Table (SLS Combinations)

| Beam | Max Disp mm | Location m | L/C | L/Displ | Global X mm | Global Y mm | Global Z mm |
|------|-------------|------------|-----|---------|-------------|-------------|-------------|
| 17   | 0.960       | 0.612      | 15  | 1093    | 0.008       | -3.443      | 0.000       |
| 17   | 0.867       | 0.612      | 9   | 1210    | 0.006       | -3.323      | 0.000       |
| 17   | 0.866       | 0.612      | 13  | 1211    | 0.007       | -3.358      | 0.000       |
| 17   | 0.866       | 0.612      | 14  | 1212    | 0.005       | -3.358      | 0.000       |
| 18   | 0.793       | 0.612      | 12  | 1323    | 0.016       | -3.265      | 0.000       |
| 18   | 0.793       | 0.612      | 8   | 1324    | 0.006       | -3.253      | 0.000       |
| 18   | 0.784       | 0.612      | 7   | 1339    | 0.005       | -3.243      | 0.000       |
| 17   | 0.774       | 0.612      | 12  | 1356    | 0.015       | -3.249      | 0.000       |
| 17   | 0.774       | 0.612      | 7   | 1357    | 0.006       | -3.238      | 0.000       |
| 17   | 0.773       | 0.612      | 8   | 1357    | 0.003       | -3.237      | 0.000       |
| 18   | 0.708       | 0.700      | 14  | 1483    | 0.002       | -3.540      | 0.000       |
| 18   | 0.700       | 0.700      | 13  | 1499    | 0.001       | -3.531      | 0.000       |
| 18   | 0.682       | 0.700      | 9   | 1538    | 0.002       | -3.473      | -0.000      |
| 18   | 0.606       | 0.700      | 15  | 1732    | -0.002      | -3.427      | 0.000       |
| 13   | 0.573       | 0.800      | 12  | 2095    | 0.005       | -3.586      | -0.009      |
| 13   | 0.563       | 0.800      | 8   | 2133    | 0.004       | -3.566      | -0.006      |

Max = 0.96 mm (LC 15) | L = Member 17 length (1050 mm) | Allowable  $L/100 = 10.5 \text{ mm}$  → PASS



## Horizontal Deflection Check

| Parameter                                                                      | Value  | Allowable                           | Check |
|--------------------------------------------------------------------------------|--------|-------------------------------------|-------|
| Max horizontal displacement - LC 15<br>Member 37   Member length L = 1000.0 mm | 0.5 mm | $L/200 = 1000/200 = 5.0 \text{ mm}$ | PASS  |

## Horizontal Displacement Table (SLS Combinations)

| Beam | Max Disp mm | Location m | L/C | L/Displ | Global X mm | Global Y mm | Global Z mm |
|------|-------------|------------|-----|---------|-------------|-------------|-------------|
| 37   | 0.500       | 0.500      | 15  | 1999    | 0.013       | -0.031      | 2.449       |
| 37   | 0.430       | 0.583      | 13  | 2324    | 0.016       | -0.028      | 1.984       |
| 37   | 0.429       | 0.583      | 14  | 2331    | 0.024       | -0.027      | 1.976       |
| 37   | 0.327       | 0.583      | 9   | 3059    | 0.018       | -0.027      | 1.507       |
| 52   | 0.287       | 0.667      | 15  | 3481    | 0.002       | -0.007      | 0.434       |
| 25   | 0.276       | 0.208      | 15  | 1813    | -0.004      | -0.070      | -0.229      |
| 37   | 0.260       | 0.583      | 7   | 3839    | 0.023       | -0.026      | 1.009       |
| 37   | 0.259       | 0.583      | 8   | 3858    | 0.031       | -0.024      | 1.001       |
| 37   | 0.259       | 0.583      | 12  | 3862    | 0.061       | -0.020      | 0.995       |
| 25   | 0.257       | 0.208      | 13  | 1943    | -0.004      | -0.069      | -0.416      |
| 25   | 0.255       | 0.208      | 14  | 1960    | 0.002       | -0.070      | -0.436      |
| 25   | 0.249       | 0.208      | 9   | 2010    | -0.003      | -0.070      | -0.390      |
| 55   | 0.235       | 0.500      | 15  | 4248    | 0.011       | -0.057      | 1.687       |
| 52   | 0.234       | 0.750      | 13  | 4277    | 0.001       | -0.005      | 0.233       |

Max = 0.5 mm (LC 15) | L = Member 37 length (1000 mm) | Allowable  $L/200 = 5.0 \text{ mm}$  → PASS

## CONCLUSION

The structural analysis and design of the **Hanging** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.453** at Member 18). Vertical and horizontal deflections are within their respective allowable limits of  $L/100 = 10.5\text{mm}$  and  $L/200 = 5.0\text{mm}$ . All scaffold tube coupler connections are adequate for the applied axial loads.

## RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

## INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Hanging scaffold configuration matches the approved drawing and is suitable for Leak Detection PoC camera installation..
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Verify all suspension points, support steelwork, clamps, anchors, and hangers against the approved drawing before any scaffold load is applied.

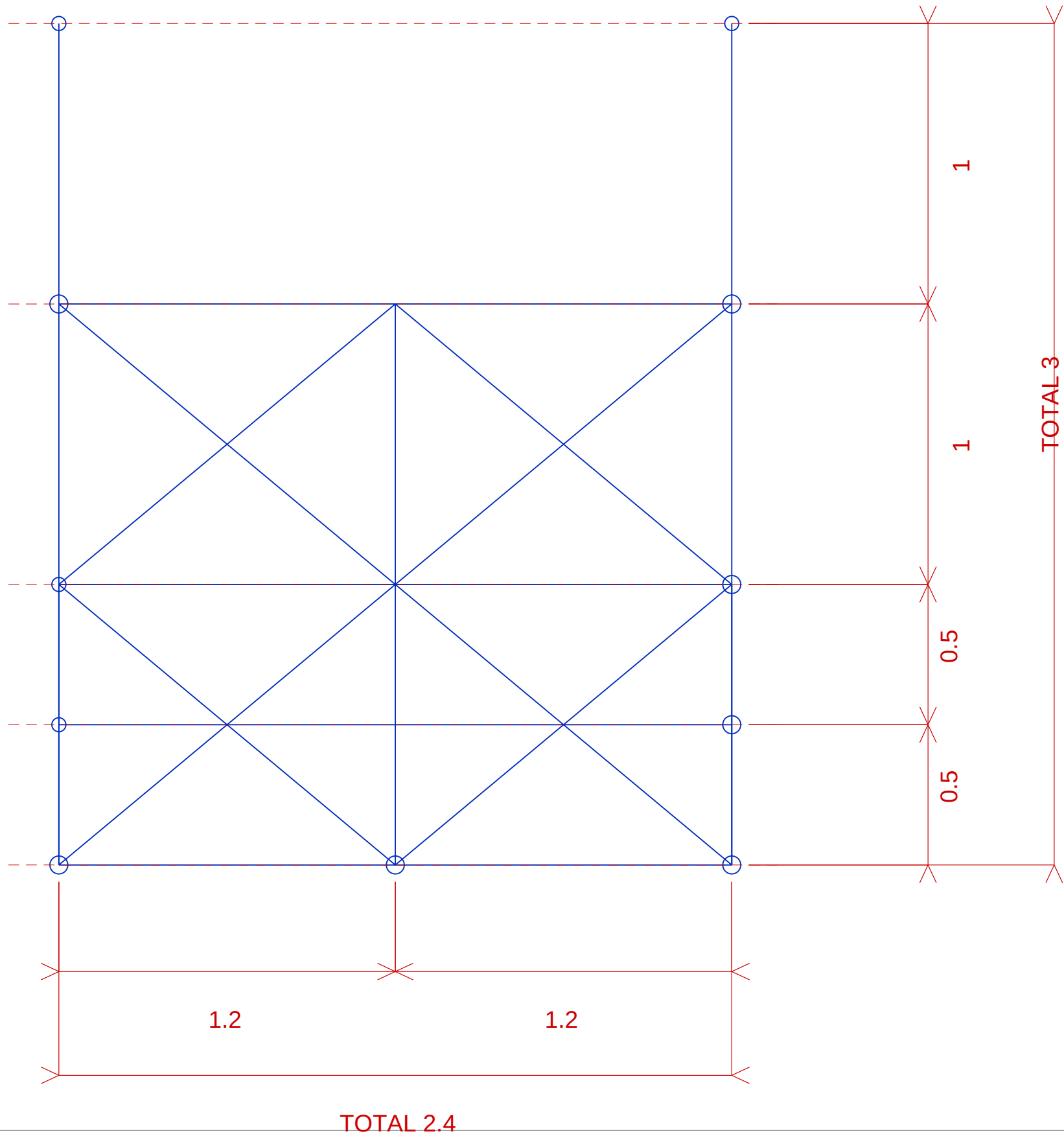
8. Assemble the hanging scaffold support frame and platform zone to the approximate overall envelope of 2.4m x 2.1m x 3.0m shown on the drawing.
9. Install hangers, ledgers, transoms, platform boards, guardrails, toe boards, and bracing progressively from the secured suspension/support points.
10. Do not load the platform until all suspension connections and secondary restraints have been inspected and signed off.
11. Install mid-level horizontal ledgers and transoms at 1.0m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
13. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
14. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
15. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
16. Gravlock or box-tie the scaffold structure to the existing structure at all indicated tie points at 0.0m height as shown on the working drawing.
17. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
18. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
19. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer.  
End-user or third-party modification of this scaffold structure is strictly prohibited.
20. For dismantling, perform the erection steps in reverse order.

## REFERENCES & STANDARDS

| Reference           | Title                                                                                                                                  | Year |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|------|
| BS EN 12811-1:2003  | Temporary Works Equipment - Scaffolds - Performance Requirements and General Design                                                    | 2003 |
| BS EN 1993-1-1:2005 | Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings                                                         | 2005 |
| BS EN 1991-1-4:2005 | Eurocode 1: Actions on Structures - Wind Actions                                                                                       | 2005 |
| BS EN 1990:2002     | Eurocode: Basis of Structural Design — Load Combinations, Partial Factors, and Limit States (ULS/SLS/EQU)                              | 2002 |
| NASC TG20:13        | Guide to Good Practice for Scaffolding with Tubes and Fittings                                                                         | 2013 |
| Durst, C.S.         | Wind Speeds Over Short Periods of Time. The Meteorological Magazine, 89, 181–186 — basis of 3-second gust to 10-minute mean conversion | 1960 |
| STAAD.Pro CONNECT   | STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software                                                                | 2022 |

**Software Used:** STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

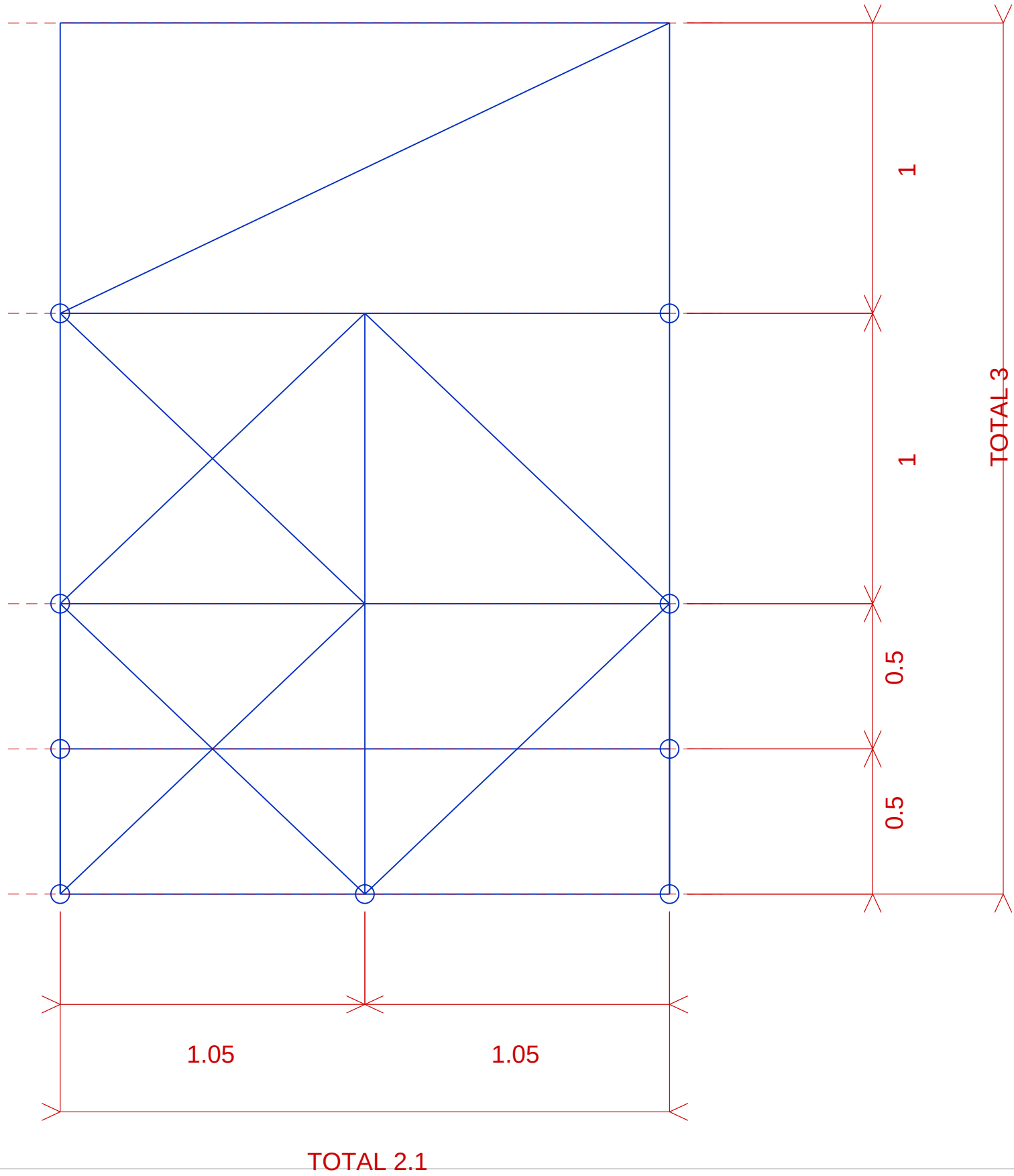
**Report:** Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



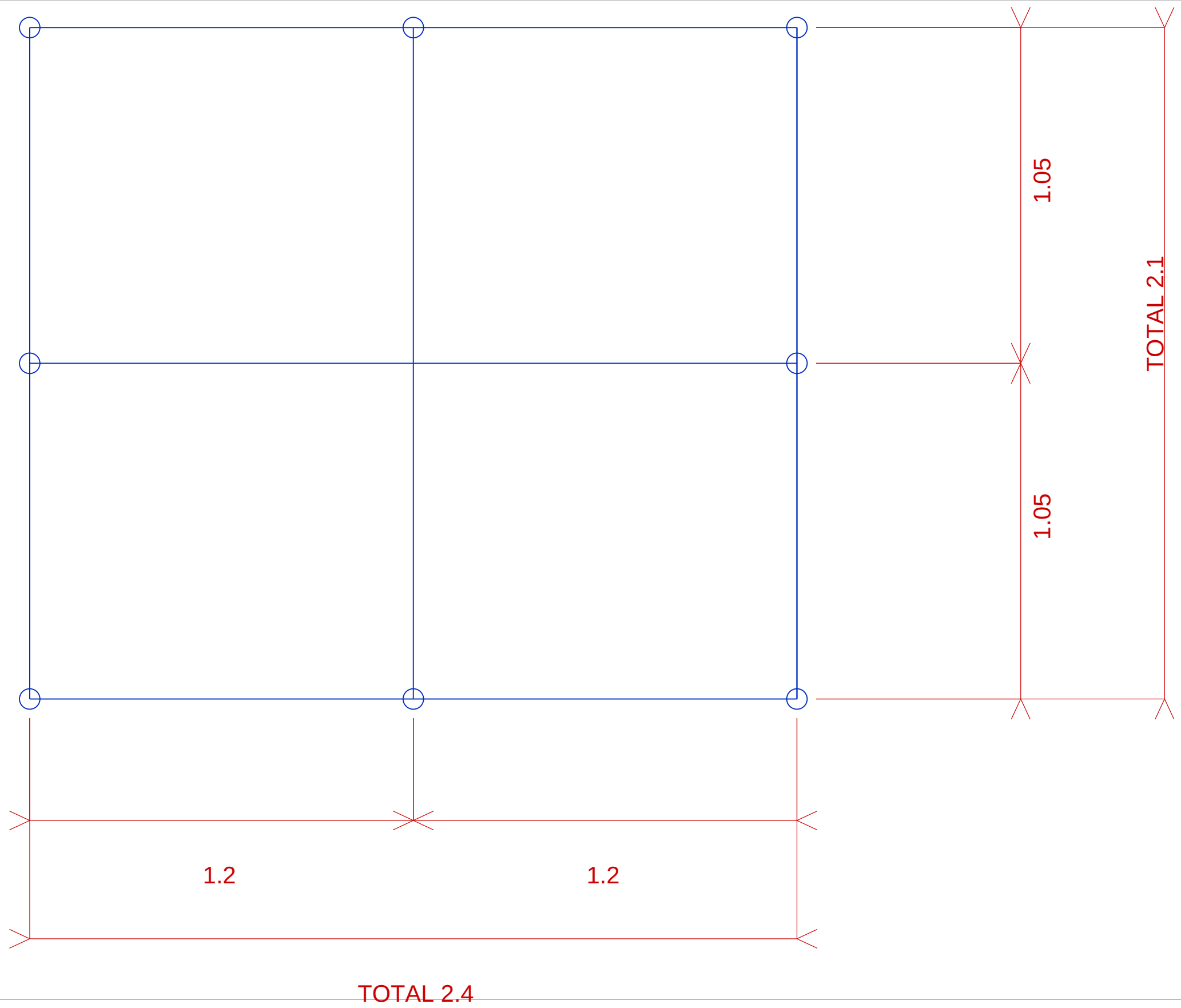
TOTAL 2.4

FRONT VIEW

|                                               |          |                                           |                             |                             |                                                                                                  |                                                                                       |
|-----------------------------------------------|----------|-------------------------------------------|-----------------------------|-----------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| PROJECT<br>SCAFFOLDING WORKS                  |          | DRAWING NO.<br>PMS-WA-ARCO-042-FRONT-VIEW |                             |                             | <br>COMPANY |  |
| DRAWING TITLE<br>PMS-WA-ARCO-042 - FRONT VIEW |          | SCALE<br>NTS                              | DATE<br>06-JUL-2026         | REV<br>0                    |                                                                                                  |                                                                                       |
| CLIENT                                        | LOCATION | DRAWN<br>Nnamani, Ifeanyi                 | REVIEWED<br>Ezewoko, Joseph | APPROVED<br>Chima, Tasieobi | SHEET<br>1 OF 4                                                                                  | SIZE<br>A3                                                                            |

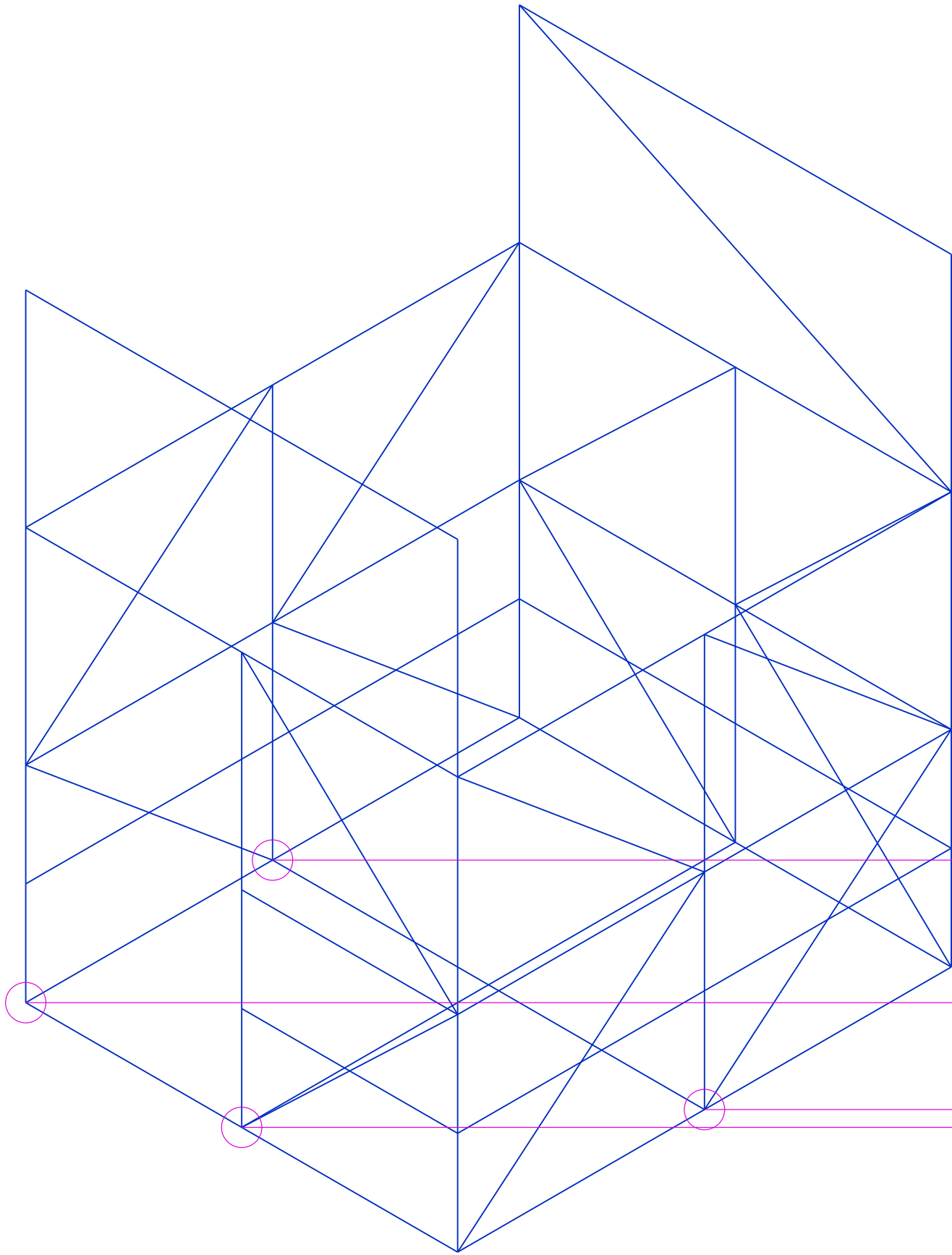


|                                              |          |                                          |                             |                             |                                                                                                                                                                             |
|----------------------------------------------|----------|------------------------------------------|-----------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROJECT<br>SCAFFOLDING WORKS                 |          | DRAWING NO.<br>PMS-WA-ARCO-042-SIDE-VIEW |                             |                             |   |
| DRAWING TITLE<br>PMS-WA-ARCO-042 - SIDE VIEW |          | SCALE<br>NTS                             | DATE<br>06-JUL-2026         | REV<br>0                    |                                                                                                                                                                             |
| CLIENT                                       | LOCATION | DRAWN<br>Nnamani, Ifeanyi                | REVIEWED<br>Ezewoko, Joseph | APPROVED<br>Chima, Tasieobi | COMPANY                                                                                                                                                                     |
|                                              |          |                                          |                             |                             | SHEET<br>2 OF 4                                                                                                                                                             |
|                                              |          |                                          |                             |                             | SIZE<br>A3                                                                                                                                                                  |



PLAN VIEW

|                                              |          |                                          |                             |                             |                                                                                                  |                                                                                       |
|----------------------------------------------|----------|------------------------------------------|-----------------------------|-----------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| PROJECT<br>SCAFFOLDING WORKS                 |          | DRAWING NO.<br>PMS-WA-ARCO-042-PLAN-VIEW |                             |                             | <br>COMPANY |  |
| DRAWING TITLE<br>PMS-WA-ARCO-042 - PLAN VIEW |          | SCALE<br>NTS                             | DATE<br>06-JUL-2026         | REV<br>0                    |                                                                                                  |                                                                                       |
| CLIENT                                       | LOCATION | DRAWN<br>Nnamani, Ifeanyi                | REVIEWED<br>Ezewoko, Joseph | APPROVED<br>Chima, Tasieobi | SHEET<br>3 OF 4                                                                                  | SIZE<br>A3                                                                            |



Tie points to  
existing structures  
at indicated points.

3D VIEW

|                                            |          |                                        |                             |                             |                                                                                                                                                                             |
|--------------------------------------------|----------|----------------------------------------|-----------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROJECT<br>SCAFFOLDING WORKS               |          | DRAWING NO.<br>PMS-WA-ARCO-042-3D-VIEW |                             |                             |   |
| DRAWING TITLE<br>PMS-WA-ARCO-042 - 3D VIEW |          | SCALE<br>NTS                           | DATE<br>06-JUL-2026         | REV<br>0                    |                                                                                                                                                                             |
| CLIENT                                     | LOCATION | DRAWN<br>Nnamani, Ifeanyi              | REVIEWED<br>Ezewoko, Joseph | APPROVED<br>Chima, Tasieobi | COMPANY                                                                                                                                                                     |
|                                            |          |                                        |                             |                             | SHEET<br>4 OF 4                                                                                                                                                             |